🏛️ GovFix: Campus Workflow Engine

Problem Understanding, Solution Proposal & System Architecture Report

VORTEXA 3.0 Hackathon
Domain 7: Government Systems & GovTech
Selected Track: **University Service Requests**
Team: **OrbitX** | Oct 11, 2026

1. Problem Understanding

Higher education campuses—such as **Dr. D. Y. Patil Institute of Technology, Pimpri**—operate as dense micro-cities housing thousands of students, researchers, faculty, and administrative staff. Every semester generates thousands of critical service requests across:

- **Campus Facilities & Infrastructure**: Laboratory electrical failures, classroom smart-board/projector breakdowns, hostel plumbing and water maintenance.
- **Academic IT & Computing Services**: Department Wi-Fi access point downtime, computing lab server faults, student portal credential blocks.
- **Administrative & Equipment Requests**: Classroom reallocation, asset relocations, and laboratory equipment calibration clearances.

Despite campus ERP portals (e.g., Samarth eGov, Google Forms, or basic email ticketing), request lifecycles systematically collapse due to **five structural failure modes**:

1. The "Limbo" Problem
Tickets enter "In Progress" and sit for weeks with zero progression forcing or technician accountability.

2. Premature & Ghost Closures
Technicians mark tasks "Closed" without physical repair photos, supervisor sign-offs, or student confirmation.

3. Silent SLA Breaches
Deadlines pass unnoticed because campus databases lack background daemons to notify Estate Managers or HODs.

4. The Flaky AI Trap
Unconstrained LLM chatbots hallucinate status jumps, approve unauthorized closures, or drop records when given mutation rights.

> **Critical Failure: Relational database rows in university ERPs are mutable. If a high-voltage lab hazard or server failure is neglected, standard databases cannot cryptographically prove when deadlines were missed or if logs were retroactively edited.**

2. Proposed Solution: GovFix (University Track)

GovFix is an institutional-grade, backend-enforced finite-state workflow engine purpose-built for university campus maintenance and IT service operations, governed by one ironclad rule:

> **"Deterministic Rules Govern Authority, State Transitions & Deadlines; AI Informs But Never Overrides."**

GovFix replaces ad-hoc ticketing with five tightly engineered subsystems:

- **Deterministic Finite State Machine (FSM)**: Formally defines the exact lifecycle: `Draft → Submitted → Triaged → Assigned → Accepted → In Progress → Ready for Closure / Escalated → Closed or Reopened`. Illegal moves (e.g. `Submitted → Closed`) return `HTTP 409 Conflict` (`ERR_ILLEGAL_STATE_JUMP`).
- **Backend-Enforced Role RBAC**: Separates duties across Requesters, Department Supervisors, and Technicians. Unauthorized attempts return `HTTP 403 Forbidden` (`ERR_ROLE_UNAUTHORIZED`) and are recorded in the audit trail.
- **Campus SLA & Multi-Tier Auto-Escalation Daemon**: Assigns deadlines by urgency (Lab Hazard: 12h, IT Server Down: 24h, Routine Maintenance: 48h). Background daemons auto-escalate overdue tickets to Tier-1 (Department Supervisor) and Tier-2 (Estate Officer/Registrar).
- **Duplicate Escalation Prevention**: Suppresses redundant escalation events for the same request across evaluation cycles.
- **Tamper-Evident SHA-256 Cryptographic Audit Ledger**: Surpassing basic SQLite, every accepted state transition, refused unauthorized move, and escalation event is cryptographically hashed and chained into an immutable ledger with 1-click mathematical verification.

- **Virtual Time-Travel Simulation**: Built-in fast-forward controls (`+12h` , `+24h` , `+48h`) allow judges to witness live automated escalations in seconds.

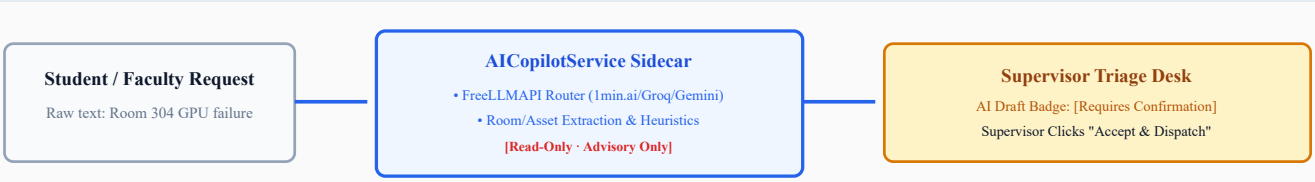
3. Intended Users & Role Boundaries

GovFix enforces strict, backend-guarded boundaries across five campus user roles:

User Role	Permitted Actions	Prohibited Actions (Strictly Refused)
Requester (Student / Faculty)	• Create request with Room & Asset ID • View own requests & live SLA countdown • Confirm closure with satisfaction score (1-5) • Reopen within 48h if issue persists	• Cannot assign technicians or officers • Cannot force-close tickets • Cannot alter SLA deadlines
Supervisor (HOD / Estate Officer)	• Review AI triage recommendations • Verify category, room, priority • Dispatch work orders to Technicians • Approve or reject closure proposals	• Cannot skip engineering repair stage • Cannot close tickets without evidence • Cannot erase historical audit records
Officer (Maintenance / IT Tech)	• Accept or reject assigned work order • Transition ticket to `In Progress` • Upload work completion notes & photo proof • Submit `Ready for Closure` proposal	• Cannot approve own work as final closed • Cannot reassign tickets to others • Cannot alter SLA timers
Campus Administrator (Registrar)	• Campus-wide SLA compliance oversight • Review Tier-1 / Tier-2 escalation rosters • Direct intervention on overdue campus hazards	• Cannot bypass state machine guards • Cannot modify cryptographic ledger blocks
System SLA Daemon	• Automated background deadline checking • Creates escalation events upon SLA breach • Enforces duplicate escalation suppression	• Cannot close or resolve tickets

4. Constrained AI Integration (FreeLLMAPI & Offline Heuristics)

In strict compliance with hackathon safety constraints: **AI operates as a non-authoritative advisory sidecar with zero state-mutation authority.**



- **Room & Asset Identification**: Automatically extracts room codes (e.g., *"DYP-MECH-W201"*) and equipment tags from request text.
- **Recurrence & Duplicate Detection**: Identifies repeated failures in the same classroom or lab within 48 hours.
- **100% Offline Resilience**: If external APIs disconnect, embedded keyword and regex heuristics take over instantly with zero downtime.

5. Differentiation From Existing Solutions

Capability / Criterion	Samarth eGov (ERP)	ManageEngine ServiceDesk	Freshservice (Cloud ITSM)	GovFix (Team OrbitX)
Workflow Core	Basic ticketing form	Hardcoded status lists	Proprietary cloud rule engine	**Deterministic 11-State FSM**
Refusal Handling	Generic validation failure	Generic UI error	Generic UI error	**Explicit HTTP 409 & 403 Refusal Codes**
Audit Ledger Integrity	Mutable SQL rows	Application database logs	Vendor cloud logs	**Tamper-Evident SHA-256 Hash Chain**
Refused Actions Logged	❌ Ignored	❌ Ignored	❌ Ignored	✅ **Immutable record with Actor, Role & Reason**
Multi-Tier Escalation	Basic email reminder	Notification trigger	Workflow trigger	**Automated Tier-1 & Tier-2 Daemon Jumps**
Duplicate Escalation Check	❌ None	❌ None	Basic throttle	✅ **State-verified duplicate suppression**
Evaluator Time-Travel	❌ None	❌ None	❌ None	✅ **Virtual Clock (+12h, +24h, +48h)**
Deployment & Cost	Heavy ERP deployment	High enterprise license	Expensive monthly SaaS	**Zero Paid APIs · 100% Local Run MVP**

6. System Architecture & Workflow Diagram

The architecture cleanly isolates user portals, authorization enforcement, workflow logic, audit immutability, and advisory AI:

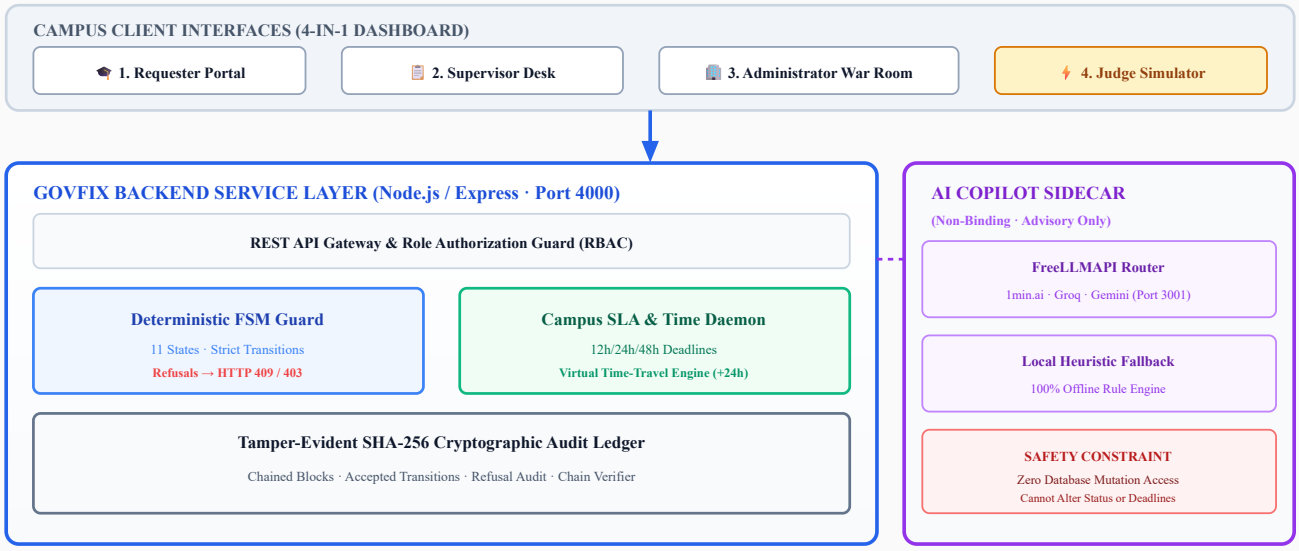


Figure 1: GovFix University Service Requests System Architecture & Workflow Engine.

7. Verification & Synthetic Testing Results

GovFix has been programmatically verified against the official benchmark evaluation suite (`npm test`), passing 100% of the functional test cases:

#	Test Name	Observed Result	Status
1	Valid Request Creation	Request created with Public ID, Room Tag, SLA deadline & audit block	PASS
2	Deterministic Refusal (Illegal Jump)	SUBMITTED → CLOSED rejected with code `ERR_ILLEGAL_STATE_JUMP` [409]	PASS

#	Test Name	Observed Result	Status
3	RBAC Authorization Refusal	Requester attempting supervisor triage rejected with `ERR_ROLE_UNAUTHORIZED` [403]	PASS
4	Supervisor Triage & Dispatch	Valid transition SUBMITTED → TRIAGED → ASSIGNED to Technician	PASS
5	Technician Acknowledgment	Technician accepts work order; transition ASSIGNED → IN_PROGRESS	PASS
6	Closure Evidence Guard	Proposing closure without completion evidence is strictly rejected	PASS
7	Virtual Time-Travel Escalation	+24h time-advance triggers automated Tier-1 escalation	PASS
8	Duplicate Escalation Suppression	Repeated overdue checks do not spam duplicate escalation events	PASS
9	Requester Resolution Confirmation	Requester confirms resolution with rating; ticket moved to CLOSED	PASS
10	Cryptographic Chain Verification	100% SHA-256 forward hashes mathematically valid	PASS

Summary: Team OrbitX has delivered a 100% locally runnable, rigorously tested, tamper-evident workflow engine ready for live evaluation at VORTEXA 3.0.